

Day-ahead forecast of heating need in university building

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ABSTRACT

This study aims to forecast day-ahead heating demand for a university building using machine learning techniques. The models utilized—logistic regression, shallow neural networks (SNN), and deep neural networks (DNN)—were trained using weather and temporal data, including variables such as outdoor temperature, wind speed, and global radiation. The objective was to compare the performance of these models in predicting space heating demand with high accuracy. The logistic regression model achieved a test accuracy of 92.03%, while the shallow and deep neural networks reached 93.39% and 93.72%, respectively. The shallow neural network demonstrated a better balance between complexity and generalization, with lower risk of overfitting compared to the deep neural network, which showed a slight drop in validation accuracy. The study highlights that while deeper networks can capture complex patterns, simpler architectures like shallow networks can offer high performance with lower computational demand. These findings contribute to optimizing energy forecasts, particularly for applications in smart building management and district heating systems.

1. Introduction

Forecasting energy demand with precision is a cornerstone of modern energy management, facilitating enhanced sustainability and operational efficiency. Accurate predictions enable energy providers and building managers to optimize energy production, distribution, and consumption, thereby minimizing costs and mitigating environmental impacts. This capability is especially critical for incorporating renewable energy sources into power grids, as their variability necessitates robust forecasting models to maintain grid stability and reduce dependence on fossil fuels. Moreover, these forecasts play a pivotal role in achieving global sustainability objectives, such as the United Nations Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action).

This study seeks to design and evaluate advanced machine learning models for forecasting heating demand in a university building. The specific goals include:

1. Examining the effectiveness of various machine learning algorithms in predicting heating demand.
2. Identifying and analyzing critical input features that influence the accuracy of heating demand forecasts.
3. Comparing the performance of developed models with existing approaches in the literature to highlight both advancements and potential areas for refinement.

1.1. Literature review

Recent advancements in machine learning have revolutionized the accuracy and applicability of energy demand forecasts. By synthesizing insights from existing research, this study aims to construct a robust forecasting model tailored to the unique characteristics of the case study. The following review of key literature in Table 1 provides a comprehensive overview of methodologies, input features, and optimization strategies employed in prior works.

Table 1: Literature review providing overview of methodologies, input features, and key insights adapted in this work

Study	Algorithm	Input Features	Optimization	Data Split	Metric	Key Insight
[1]	Deep Neural Networks (DNN)	Weather parameters, building data for aging residential structures in Korea	Trial-and-error for hyperparameters (hidden layers: 1–10, neurons: 10–30)	Training (70%), validation (15%), testing (15%)	Coefficient of determination (R^2)	Demonstrated the capacity of DNNs to model intricate relationships between climatic variables and energy consumption.
[2]	Deep Neural Networks, Linear Models (ridge, lasso, polynomial regression)	Temporal patterns (hour of the day, day of the week), historical heat load, temperature forecasts	Automated feature selection and hyperparameter tuning	Training/validation (90%), testing (10%)	Root Mean Square Error (RMSE)	Highlighted the significance of temporal features for improving district heating load predictions.
[3]	Artificial Neural Networks (ANN), Linear Regression Models (LRM), k-Nearest Neighbors (k-NN)	Weather data (temperature, solar irradiation, wind speed, humidity), occupant behavior (e.g., domestic hot water consumption, occupancy patterns)	-	Training (60%), testing (40%)	RMSE	Emphasized the importance of integrating occupant behavior and meteorological data in forecasting models.
[4]	DNN, Random Forest, Gradient Boosting Machine (GBM), Histogram-GBM, k-NN	Building-specific parameters (floor area, heating degree days, construction year, heat source)	Trial-and-error, min-max, and z-score normalization	Training (80%), testing (20%)	-	Demonstrated the value of data preprocessing and normalization for enhancing model accuracy.
[5]	Deep Neural Networks (DNN)	Hourly electricity consumption data for hospital buildings	Hyperparameter tuning (neurons, hidden layers, activation functions)	Training/validation (90%), testing (10%)	Coefficient of determination ($R^2 = 95\%$)	Illustrated the scalability of DNNs in handling complex dependencies and large datasets.

The literature demonstrates the versatility and effectiveness of machine learning algorithms, particularly DNNs, in energy demand forecasting. Effective models often utilize a combination of weather data, temporal patterns, and building-specific characteristics as input features. Optimization techniques, such as hyperparameter tuning and data normalization, significantly enhance model performance. Building on these insights, this study employs DNNs and integrates critical input features and optimization strategies to achieve accurate and reliable heating demand forecasts for the selected case study.

2. Materials and Methods

2.1. Case study

The focus of this study is the U-Building at KTH in Stockholm, Sweden. This facility spans approximately 3,600 m² across seven floors, serving as a representative case for investigating energy demand forecasting in educational buildings. The geographical location in Stockholm provides a diverse climatic context, with significant seasonal temperature variations influencing heating needs. These characteristics make the U-Building an ideal candidate for evaluating advanced forecasting models.

2.2. Data collection

For the data, we used the following input features: Hour, Relative Humidity, Global Radiation, Wind Speed, Outdoor Temperature, Wind Direction, Day, Chronological Day, and Space Heating. The Table 2 below provides a brief overview of the variables used in the different case studies we reviewed and highlights which variables we decided to use from each case study based on the data we had available.

Table 2: Table highlighting the literature on variables considered in the study

Variables/Parameters	Source
Temperature data (7 days prior, hourly), hour of the day (1-24), weekday, chronological day, past hourly heat demand	[2]
Outside temperature, solar irradiation, wind speed, humidity	[3]

In the study *Data-Driven Approach to Forecast Heat Consumption of Buildings with High-Priority Weather Data* by Golmohamadi H. [3], solar irradiation and outside temperature were identified as the most important input variables for the model's predictions. We observed that many other studies also used these input variables, and given that heat transfer from a building to its surroundings is closely correlated with heat loss, their inclusion was an obvious choice. We also noticed that some studies included humidity and wind speed. However, from the original dataset, we had access to variables such as volumetric flow for building cooling and heating, as well as cooling demand. We decided to exclude these variables, as they are closely related to heat demand and would likely increase training time without adding significant complexity to improve the model. In addition to the standard input variables, we added three others. First, using the timestamp from the raw data, we created a variable ranging from 0 to 23 to represent the hour of the day. This was inspired by other case studies and enables the model to detect patterns in building usage, such as higher occupancy during certain hours. Second, we introduced a chronological day variable, which assigned each day a number based on its position in the year, with 1 representing January 1st and 365 representing December 31st. This addition, also inspired by case studies, allows the model to detect seasonal variations. Lastly, we included a weekday variable ranging from 1 to 7 to represent the day of the week. This decision was based on findings from case studies that suggested this could help the model recognize different usage patterns, particularly in buildings like schools, which are primarily used on weekdays. These adjustments ensured the model could effectively capture both temporal and seasonal patterns in building usage while remaining efficient.

2.3. Data cleaning and preparation

The data was for one whole year, and was separated in timesteps of one hour each. A summary of the data before cleaning can be viewed in Table 3. In Table 3 some outliers can quickly be spotted such as the maximum of the wind speed or the minimum in space heating.

Table 3: Summary of data before cleaning

	Relative humidity	Global radiation	Wind speed	Outdoor temperature	Wind direction	Space heating
Average	81.61	19.34	6.97	4.53	198.39	12-2
Minimum	25.55	0	0.15	-5.29	15.12	-155.92
Maximum	100	237.82	3276.7	18.24	3277	51.56

During the data preprocessing stage, outliers were addressed by removing the corresponding rows entirely rather than smoothing the data or filling in missing values. This approach was chosen because the number of rows removed due to outliers was small, and removing them had negligible effects in terms of the overall dataset's integrity. Methods such as rolling averages or exponential smoothing, while they could be applied, could introduce bias or inaccuracies, which we sought to avoid. Furthermore, since we did not analyze the dataset as a time series, the presence of non-coherent data points was not a concern.

To focus on forecasting heating demand, the analysis was restricted to data points collected between October 1st and April 30th. Including summer months would have added unnecessary computational complexity and potentially disrupted the model's learning process. As part of the initial data cleaning process, clearly erroneous values were removed. For example, wind direction values outside the valid range of 0–360 degrees were excluded, as values like 540 degrees are nonsensical and cannot be reasonably approximated. This step led to the removal of 10 rows. Additionally, rows with negative space heating demand values were removed, as these are invalid within the context of our analysis, resulting in the exclusion of 2 rows.

Also, duplicate timestamps were identified and addressed in the dataset. Upon inspection, these duplicates represented repeated values. To resolve this, a duplicated value was replaced with its nearest counterpart. For instance, an issue occurred in rows 4322 and 4323, both associated with the timestamp "2020-03-29 03:00:00."

To enhance outlier detection, Mahalanobis distance was applied, a measure that accounts for the multivariate mean, correlation, and covariance of variables. Unlike Euclidean distance, which considers deviations along individual dimensions, the Mahalanobis distance evaluates how data points deviate from the overall correlation structure of the dataset. By setting a threshold of 3.5 standard deviations ($3.5 \cdot \sigma$) data points that significantly deviated in value and correlation structure were effectively identified and excluded. This threshold was adopted based on a relevant case study with inputs and scope comparable to ours, further validating its applicability to our analysis.¹

In Table 4 the results from the cleaning can be viewed. Compared to Table 3 the maximum values of wind speed were much lower, and more realistic. Same with the space heating minimum, where it was not negative anymore.

Table 4: Summary of data after cleaning for all the data.

	Relative humidity	Global radiation	Wind speed	Outdoor temperature	Wind direction	Space heating
Average	81.61	17.38	2.14	4.49	194.93	12.14
Minimum	25.55	0	0.15	-5.29	15.12	1.26
Maximum	100	193.16	14.5	16.74	333.17	40.39

In Table 5 the amount of outliers removed from the initial cleaning and the amount of outliers removed due to Mahalanobis cleaning is stated. Most of the outliers removed were for global radiation.

Table 5: Summary of number of outliers removed from different cleaning processes.

	Result from initial cleaning	Results from Mahalanobis distance outlier removal
Variable	Outliers	Outliers
Relative humidity	0	0
Global radiation	0	51
Wind speed	0	16
Outdoor temperature	0	9
Space heating	2	30
Wind direction	10	0

In Table 6 the amount of rows for the original dataset, the amount after the initial cleaning, and the amount after the cleaning done with the Mahalanobis outlier detection were demonstrated.

Table 6: Summary of amount of rows removed from outliers.

Results of the cleaning	
Step	Number of Rows
Original dataset	5112

After initial cleaning	5100
After Mahalanobis filtering	4992

After the data was cleaned and prepared the whole day was shuffled and 80% was to be used for training and the rest for testing.

For the normalization process the MinMaxScaler from scikit-learn was utilized, which takes an input variable and normalizes it between the 0 to 1 range. The normalization process involves scaling each value by subtracting the minimum value of the feature and then dividing the result by the range of the feature, which is the difference between the maximum and minimum values. This adjusts all values to fall proportionally within a range from 0 to 1.

2.4. Forecasting models (400 words)

The forecasting will be done using three methods with increased complexity. The methods in order of complexity are: logistic regression, shallow neural network and deep neural network. These methods are quite similar to each other in the overall logic of how they work. Therefore, logistic regression will first be explained, followed by the other two, adding what is new. A visual depiction of logistic regression is shown in Figure 1.

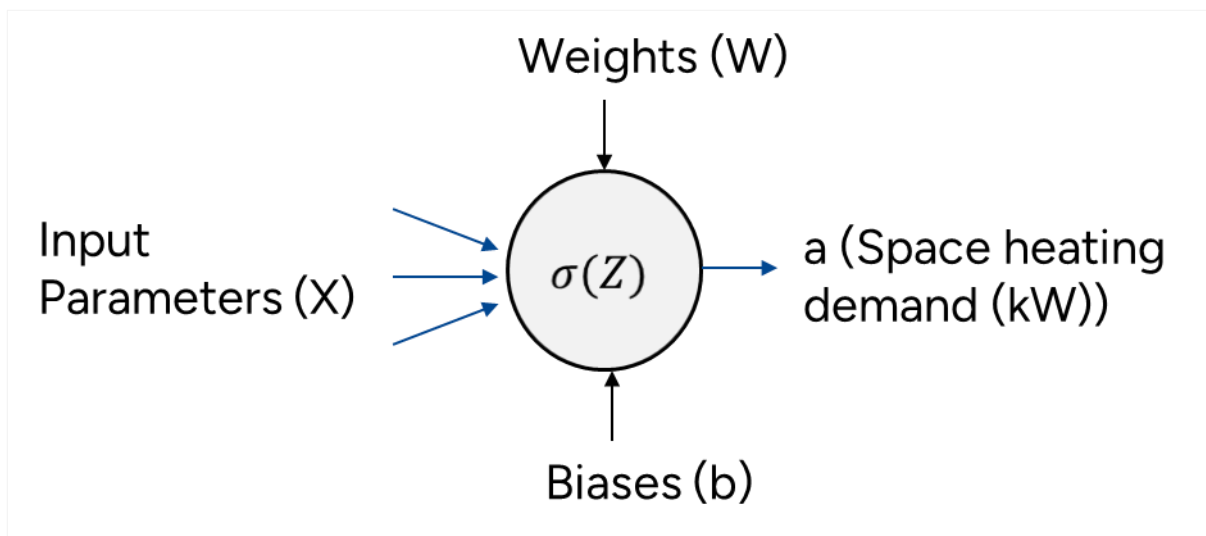


Figure 1: Logistic regression visualization

All these three models start with the data preparation and pre-processing described in the previous sections. After this, all parameters are divided into either an input feature, or an output. In our case, the output was the space heating demand (kW), and all other parameters were used as input features. Input features are all part of an input vector called X, as seen in Figure 1. All the models use weights and biases, that are updated over n iterations, where the prediction gets better with the updated weights and biases. This process has five steps. The first step is that the weights and biases are initialized. For logistic regression, it is often sufficient to initialize the weight and biases to 0. The second step is forward propagation. In forward propagation, the activation function is applied to a vector(Z) with the weights and biases. For logistic regression, the activation function is the sigmoid function.

$$Z = W^T X + b$$

$$\text{Equation (1)}$$

The activation function is applied to Z, which gives the current prediction. In the third step, the current prediction is compared with the actual values for the output parameter. This is done with the loss function.

$$\text{Loss function}(J) = -(y \log(a) + (1 - y) \log(1 - a)) \quad \text{Equation (2)}$$

From the loss function, the cost is calculated by dividing with the number of training examples. This value gives an indication of how good the prediction is. The next step is backward propagation, where gradients of the loss function with respect to the model's weights and biases are calculated. The last step is to update the weights and biases according to equation (3) and (4).

$$w = w - \alpha \frac{\partial J}{\partial w} \quad \text{Equation (3)}$$

$$b = b - \alpha \frac{\partial J}{\partial b} \quad \text{Equation (4)}$$

Shallow neural networks expand upon logistic regression by introducing one or more hidden layers between the input and output. Each layer uses weights, biases, and activation functions, such as ReLU or tanh, enabling the model to capture more complex patterns. A representation of shallow neural network is shown in Figure 2.

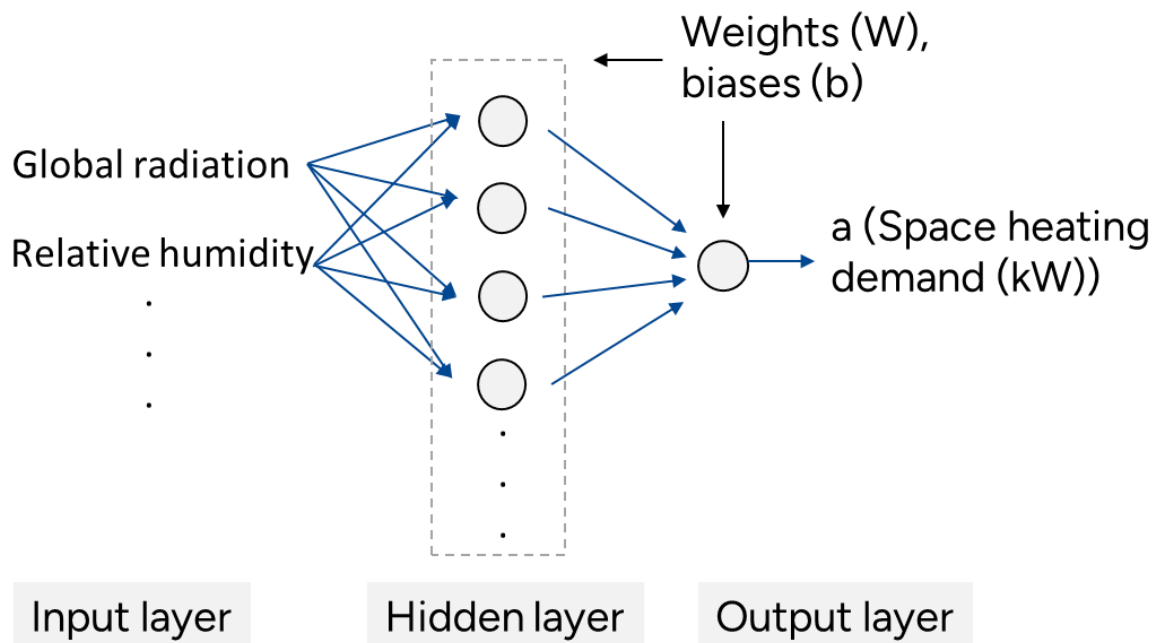


Figure 2: Representation of shallow neural network

Deep neural networks further increase complexity by adding multiple hidden layers, allowing hierarchical feature extraction and improved performance for non-linear relationships. Both methods use backpropagation and gradient descent for optimization. A pictorial representation of deep neural network is shown in Figure 3.

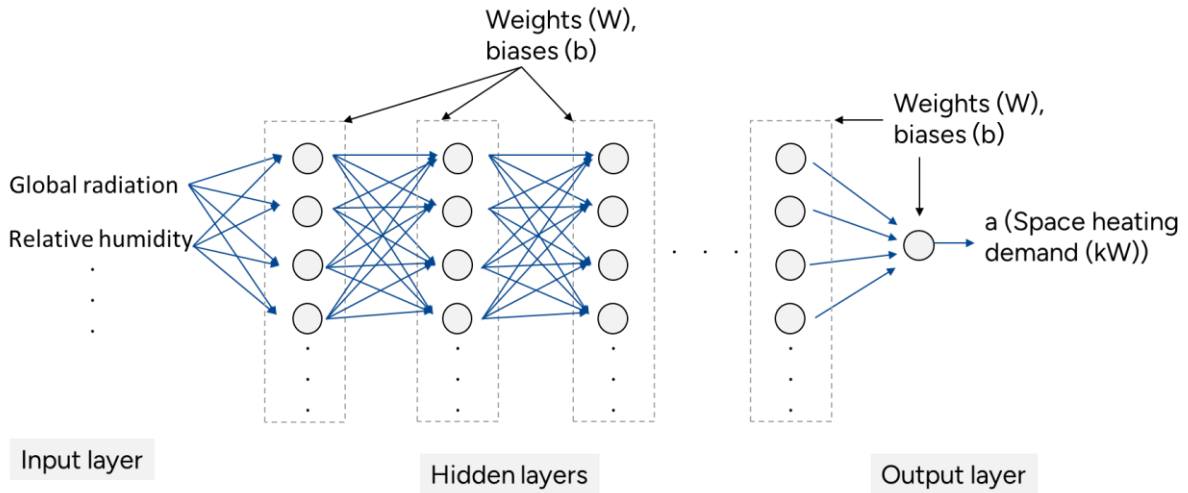


Figure 3: Representation of deep neural network

In our application of these methods, we have made two assumptions. The first one is that we did not optimize the activation function and therefore only used ReLU for all hidden layers, and sigmoid in the output layer. The second assumption was that the number of nodes were the same in all layers for the deep neural network. These assumptions were made to drastically reduce the number of possible combinations for the hyperparameters.

Table 7 shows the hyperparameter ranges that have been considered for the three models for trial and error to find the optimal hyperparameters.

Table 7: Ranges of Hyperparameters examined during trial and error to find the optimal value

	Learning Rate	Number of Iterations	Nodes in Hidden Layer	Number of hidden layers
Logistic regression	0.1-1 (in step of 0.1)	1000,2000,3000	-	-
Shallow neural network	0.1-1 (in step of 0.1)	1000,2000,3000	1 – 10 (in step of 1) [6]	-
Deep neural network	0.1-1 (in step of 0.1)	1000,2000,3000	1 – 10 (in step of 1) [6]	2 – 10 (in step of 1) [1]

3. Results

For each method, optimal values of hyperparameters were found by going through all the hyperparameters within the range mentioned above. The results with the optimal value of the hyperparameters are presented in this section.

The results of the model comparison highlight important differences in accuracy and complexity. The logistic regression model, while simple, achieved an impressive test accuracy of 92.03%. This efficient model proved effective in predicting heating demand with minimal variance between test and validation accuracy (91.96%), indicating a strong ability to generalize to new data. However, both the shallow and deep neural networks performed better in terms of test accuracy. Figure 4 and Figure 5 shows the plot of actual and values for test and validation data, respectively, simulated by the model.

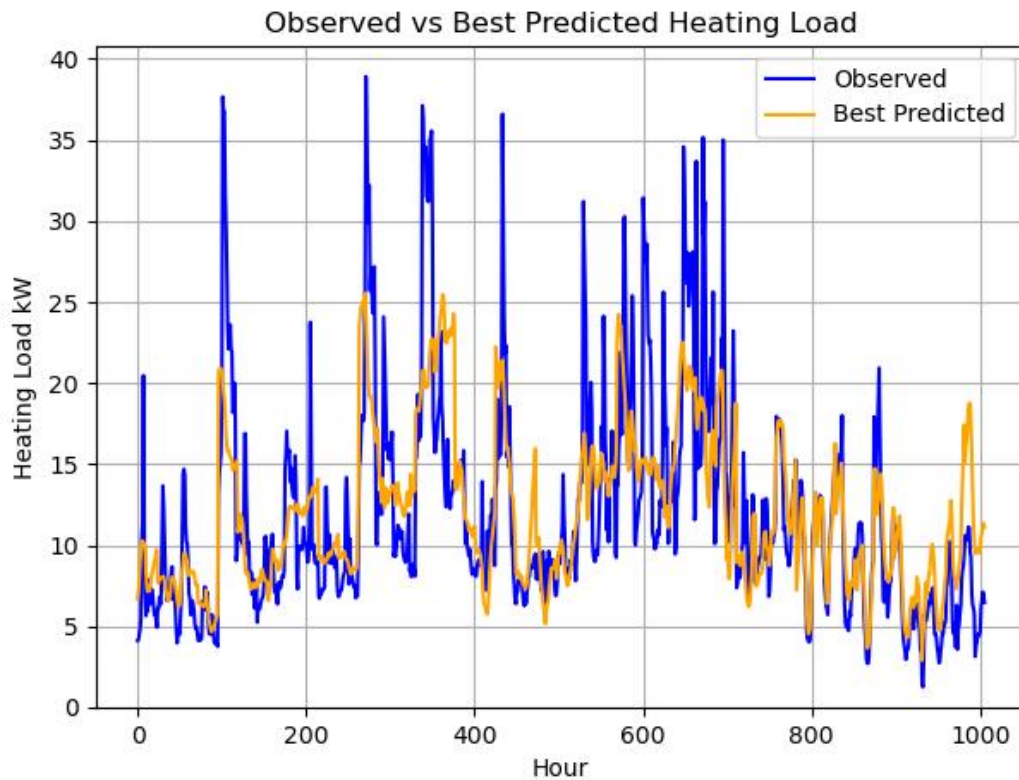


Figure 4: Graph showing observed and simulated values for test data. The model is based on logistic regression using optimal hyperparameter values.

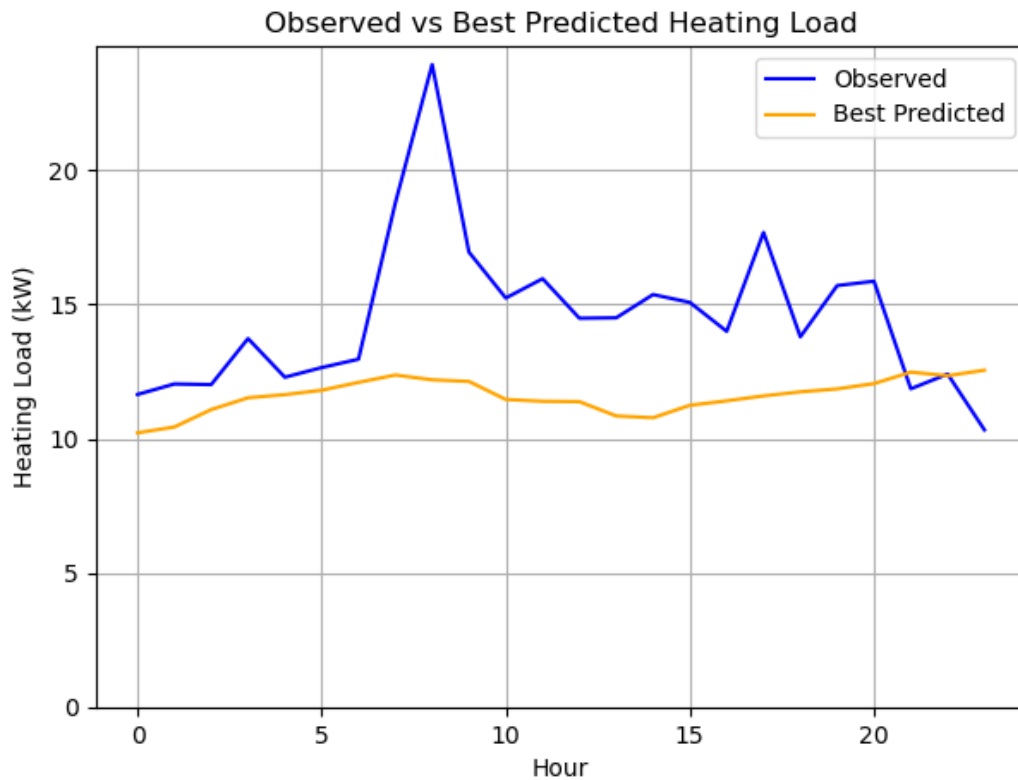


Figure 5: Graph showing observed and simulated values for validation data - January 12, 2023. The model is based on logistic regression using optimal hyperparameter values.

The shallow neural network (SNN), consisting of a single hidden layer and 4 neurons, achieved a higher test accuracy of 93.39%. The additional complexity of this model enabled it to capture more nuanced relationships within the dataset. Its validation accuracy (92.37%) closely aligned with its test accuracy, suggesting minimal overfitting and reliable generalization. This result highlights that even with relatively simple architectures, neural networks can outperform traditional models like logistic regression when forecasting heating demand. Figure 6 and Figure 7 shows the plot of actual and values for test and validation data, respectively, simulated by the SNN.

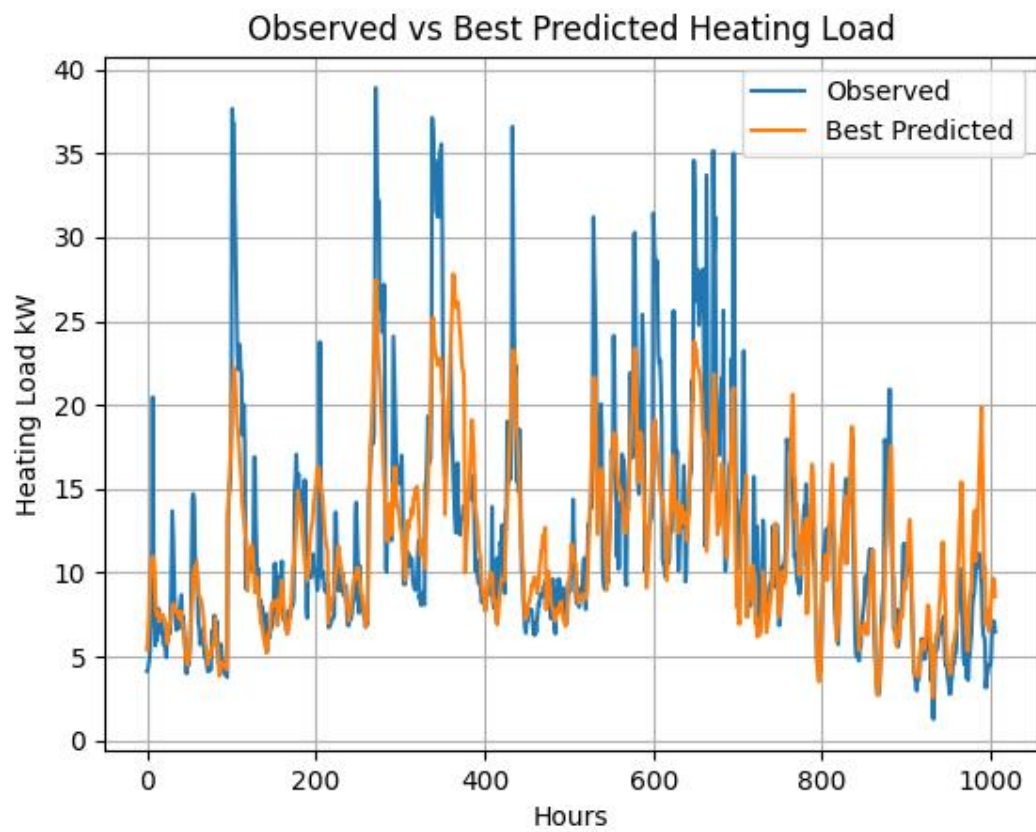


Figure 6: Graph showing observed and simulated values for test data. The model is based on shallow neural network using optimal hyperparameter values.

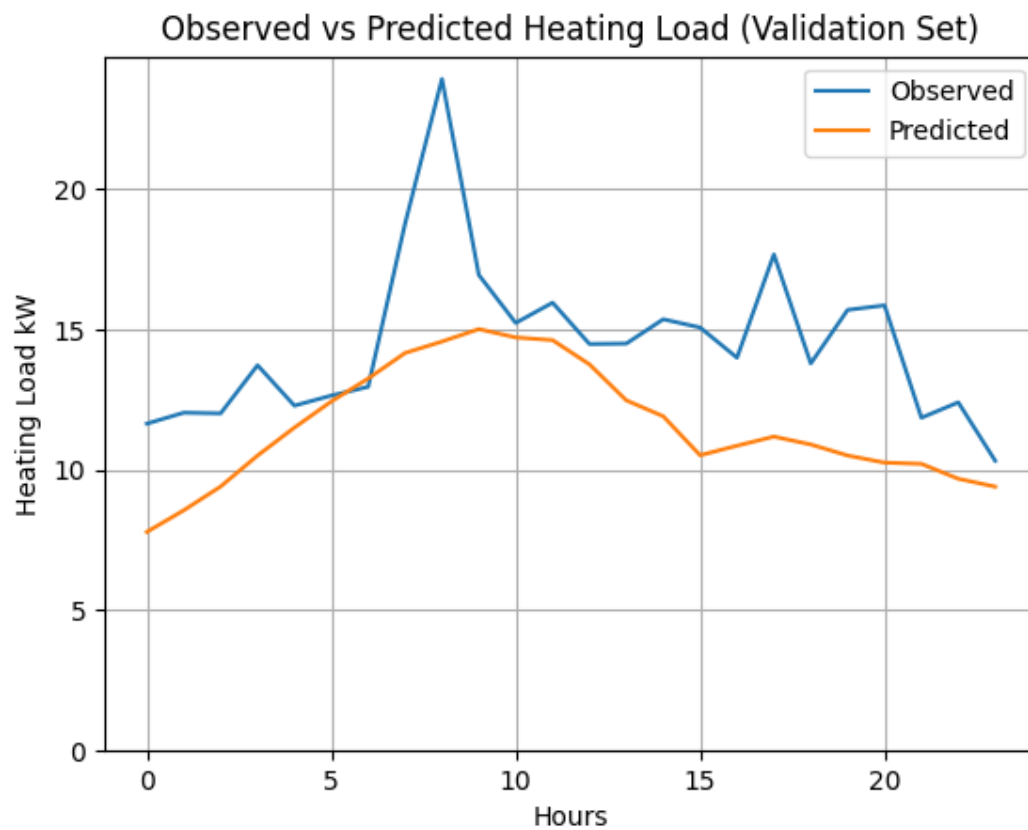


Figure 7: Graph showing observed and simulated values for validation data - January 12, 2023. The model is based on shallow neural network using optimal hyperparameter values.

The deep neural network, featuring 6 hidden layers with 8 neurons each, delivered the highest test accuracy of 93.72%. The plots in Figure 8 and Figure 9 show the model simulated values in comparison to test and validation data, respectively. A deeper network likely contributed to its ability to learn complex patterns in the data. However, a slight drop in validation accuracy to 91.53% suggests potential overfitting in the model's training process. It could also be related to uncertainties in the validation. While this deep neural network outperformed the others in terms of test accuracy, the shallow neural network demonstrated a more balanced performance between test and validation.

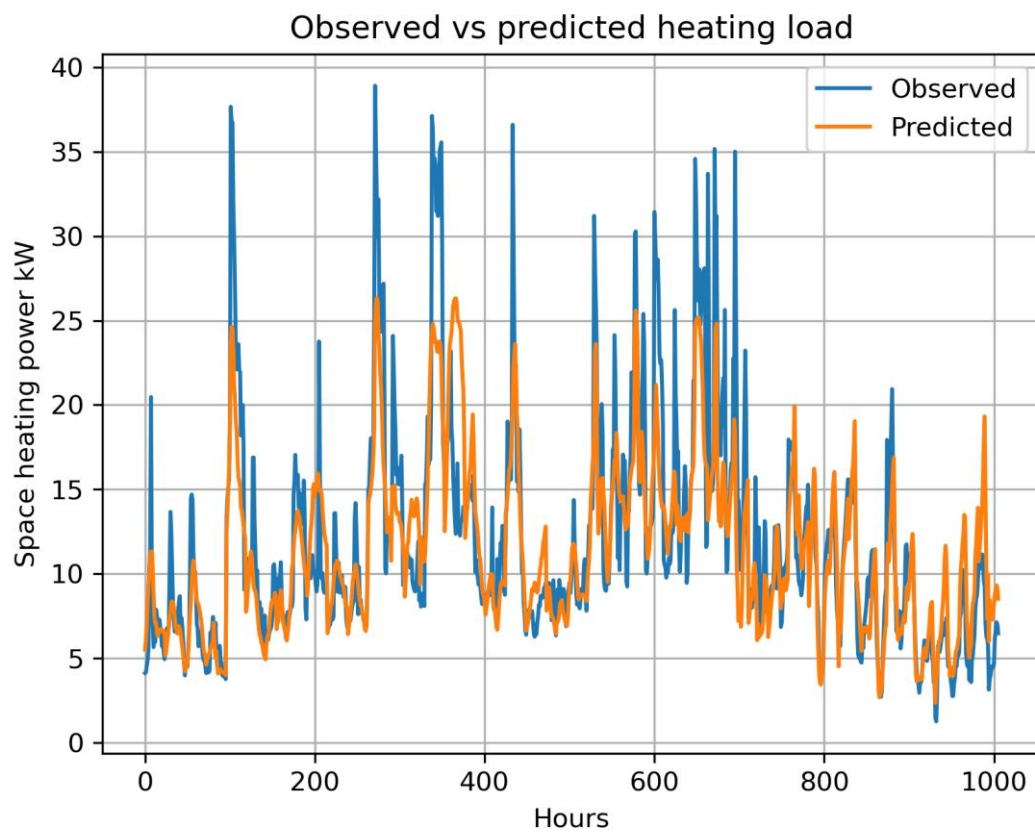


Figure 8: Graph showing observed and simulated values for test data. The model is based on deep neural network using optimal hyperparameter values.

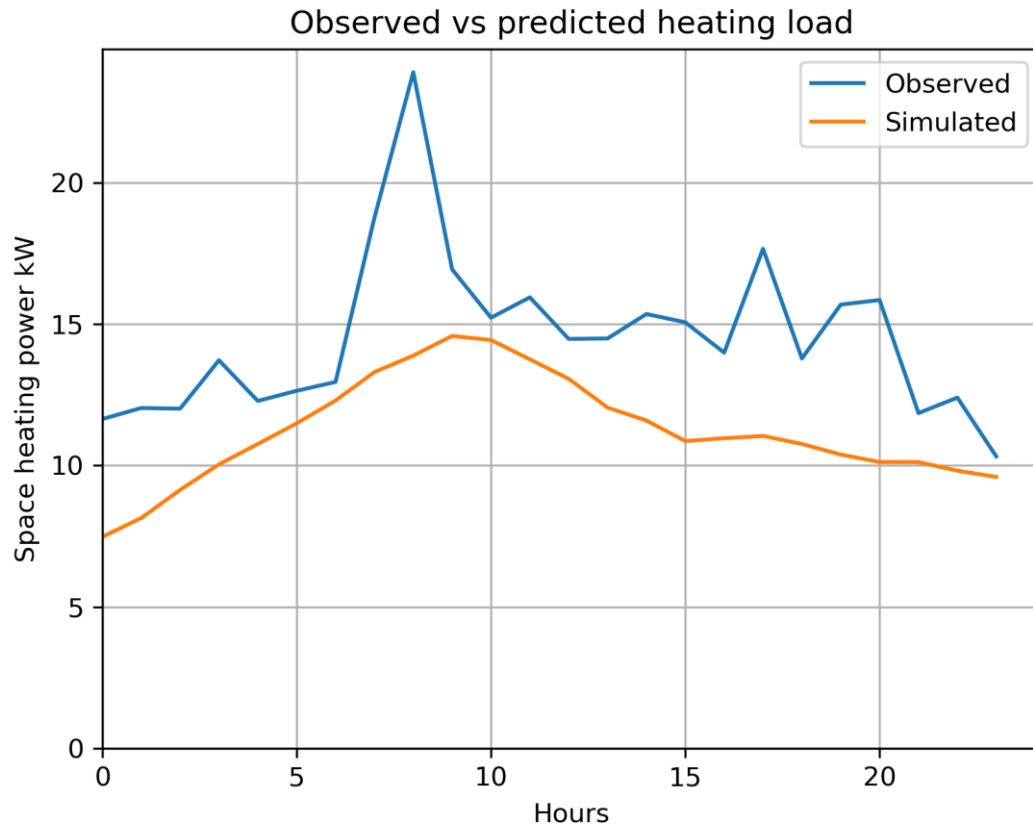


Figure 9: Graph showing observed and simulated values for validation data - January 12, 2023. The model is based on deep neural network using optimal hyperparameter values.

Table 8 shows the runtime of an iteration in each prediction model. The runtime can be used a proxy for the energy demand of each prediction and allows us to compare and justify the changes in runtime (energy) with the increase in accuracy. The run time for DNN corresponds to optimal hyperparameters. Test accuracy increased by 0.3% when a more complex DNN is used instead of SNN. The increase in accuracy is insignificant considering 10 times higher runtime for DNN relative to SNN. From a practical and climate point of view, SNN outperforms DNN in terms of performance.

Table 8: Runtime for training each model

Model	Runtime – training data (s)
Logistic regression	$8.6 \cdot 10^{-5}$
Shallow neural network	0.69
Deep neural network	7

Overall, the shallow neural network model offered a strong balance between model complexity and performance, indicating that even a modest neural architecture can yield highly accurate results for this task.

4. Discussion

4.1. Comparison with literature

This section provides a comparative analysis of the forecasting models developed in this study against those reported in the literature, highlighting the performance of each model. In this study, logistic regression was used to forecast heating demand. However, due to the absence of similar models in the literature, comparisons were made with linear regression models instead. The logistic regression model achieved a test accuracy of 92.03%, outperforming some linear regression models, such as one that reached 81.99%, but slightly underperforming others, such as the model with 93.36% accuracy in forecasting heating and cooling loads in residential buildings.

The performance of the shallow neural network (NN) models was consistent with findings from the literature. The shallow NN achieved 93.48% accuracy with 3,000 iterations and one hidden layer, with a slight improvement to 93.60% when fewer iterations and a lower learning rate were applied. These results are in line with other studies, such as predictions of heating energy in old houses, which reported an accuracy of 96.10% using deep NNs. A deep neural network with six hidden layers in this project yielded an accuracy of 93.72%, closely aligning with the results from similar studies. Overall, neural networks outperformed linear models, demonstrating their suitability for predicting heating demand.

In this study, the mean absolute error (MAE) was chosen as the primary metric for evaluating model accuracy, whereas much of the literature reports model performance using the coefficient of determination, R^2 . This methodological difference partially explains the observed variations in how the models handle extreme values. In using MAE as the performance metric, there is a possibility that the large error as seen in the prediction of extreme values is overshadowed by a higher number of relatively smaller errors in the non-extremities. In our case, this trade-off is evident in the model's difficulty in capturing the highest peaks, leading to lower predicted heating loads during these extreme conditions. Using R^2 as the performance metric in optimizing hyperparameters could be an option to capture this variation.

A valuable addition to the analysis would be a comparison of the running time required for optimizing each model. Evaluating computational efficiency is crucial, especially when working with complex models like neural networks, which often demand more processing power and time for hyperparameter tuning. Unfortunately, this aspect was not discussed in the literature reviewed and could therefore not be compared.

4.2 Concerns and biases related to AI-based energy forecasts

AI-based energy forecasts present several challenges and biases that need careful consideration. Data bias is one major concern, as models rely heavily on historical data that may reflect outdated or discriminatory patterns, leading to skewed predictions. Such biases can impact decision-making processes and reinforce existing inequalities in energy distribution and access. Model transparency and accountability are also significant issues. Many AI models, especially neural networks, operate as "black boxes," making it difficult for users to interpret how predictions are generated. This lack of visibility can undermine trust and limit the adoption of AI-based forecasts, particularly in critical applications like energy management. Human biases can further exacerbate these issues. Users may assume that AI forecasts developed for one scenario will work in other contexts with different behaviors or variables, leading to errors and reduced performance. These are some of the concerns related to AI-based forecasts.

5. Conclusion

The current study reports day-ahead energy demand for space heating of a university building using logistic regression, shallow neural network, and deep neural network. The models were able to predict the demand for space heating with a relatively high accuracy (>90%) using 7 input variables

representing temporal and weather data. Deep neural networks performed better in training and testing relative to SNN and logistic regression. However, in day-ahead forecasting, SNN showed a higher accuracy which we attribute partly to overfitting in DNN. Testing accuracy increased with the number of iterations at a constant learning rate and other hyperparameters, unless the model converged at lower iterations. We additionally observed a higher accuracy at wider (more neurons) relative to a deeper (more layers) network. Although the accuracy as measured by mean absolute error was high, the model was unable to predict the extreme values. It could partly be because the models don't fully capture/explain the variation in the data. Other performance metrics such as coefficient of determination (R^2) to drive the optimization towards capturing full variation in the data could address the issue. Using a higher temporal resolution, or training with more examples could also help models capture the full extent of the patterns within the data. Correlation analysis for identifying independent variables as observed in various literature could isolate strong predictors of the dependent variable. Using other activation functions could help models tune the prediction specific to the study. A highly accurate and general data-driven model can provide a realistic and less computationally intensive alternative to physics-based models for estimating energy demand of buildings.

6. References

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